EB-H-JEPA: Energy-Based Hamiltonian JEPA World Models

A Failure Atlas and a Controlled Sample-Efficiency Test in Crafter

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Abstract

World models that operate in a learned latent space (JEPA; LeCun 2022) are the leading bet for sample-efficient, long-horizon model-based RL (DreamerV3, Hafner et al. 2023). A recurring hypothesis is that imposing physical structure on the latent predictor — Hamiltonian, metriplectic, or GENERIC — should make learned dynamics faithful and stable. We test this hypothesis twice, with instruments of increasing fidelity. First, on a nonlinearly lifted Lorenz-63 system, we audit five structurally distinct latent predictors under a chaos-aware protocol. The result is a failure atlas: each constraint class fails, but on a property disjoint from the one it enforces. A rigid Hamiltonian predictor freezes and cannot dissipate; a naive metriplectic predictor sits on a dead $LL^\top$ saddle with dissipation pinned at exactly zero; fixing both bugs (constant divergence-free J, nonzero R initialization) makes dissipation engage and recovers the true phase-space contraction -13.667 to $\sim 1\%$ accuracy — yet spectral alignment still cannot pin individual Lyapunov exponents. Second, and constructively, we insert the fixed-metriplectic predictor into a compact DreamerV3-style agent and measure sample efficiency on Crafter (Hafner 2021), the benchmark of the world-model group that introduced RSSM, with matched 32-bit free-bits budgets and three seeds per arm. The controlled result is a latent-stability finding with matched returns: the categorical RSSM path collapses to its prior within 2k steps (raw KL 0.31 bits, 1% of its budget; latent norm exactly the uniform-one-hot expectation), while the metriplectic latent stays informative and bounded (raw KL 34.1 bits, 106% of budget; norm 1.49) with dissipation engaged and rising. Greedy evaluation returns sit at the same floor in both arms: structure neither helps nor hurts sample efficiency at this budget, and the stability is free. All code, runs, and evidence are public; every claim traces to a JSON artifact.

1 Introduction

JEPA-style world models [12, 2, 5] and their model-based RL instantiations [9, 10, 3] share a core bet: a latent predictor learned from observations can model dynamics that stay stable and meaningful over long horizons. The predictor is the crux. DreamerV3's categorical

RSSM learns it from data alone; a growing line of work asks whether injecting physical structure — symplectic/Hamiltonian form [6, 4, 16], metriplectic/GENERIC dissipation [15, 17, 11], or learned divergence gates [7] — yields predictors that are more faithful and more sample-efficient.

This paper reports a two-part test of that hypothesis, built to be reproducible end to end.

Part 1: a failure atlas on a known chaotic system. We train five structurally distinct latent predictors on Lorenz-63 lifted into a 64-D latent chart and audit each under a chaos-aware protocol: rollout boundedness, leading Lyapunov exponent, phase-space contraction, attractor overlap, and dissipation engagement. The honest headline is negative, and more precise than "structure doesn't help": *every constraint class fails, but on a property disjoint from the one it enforces.* A rigid Hamiltonian predictor (fixed canonical J, learned H) is conservative by construction and cannot dissipate; it freezes into a quasi-periodic tube with near-zero motion. A naive metriplectic predictor learns R through $R = LL^\top$ but sits on the dead saddle where $L = 0$: R stays exactly 0, dissipation never engages. Fixing both bugs — constant divergence-free J (closing the learned-skew loophole) and nonzero L initialization (breaking the dead saddle) — makes R engage and recovers the true contraction -13.667 to $\sim 1\%$. Yet even the best arm cannot pin individual Lyapunov exponents: differentiable QR spectral shaping controls only a finite-horizon statistic, not the asymptotic Oseledets spectrum [14]. And overlap metrics reward dead models: a frozen predictor sits inside the encoded attractor and posts the best Chamfer score while being dynamically inert.

Part 2: a controlled sample-efficiency test in Crafter. A failure atlas on Lorenz tells us where structure fails in isolation; it does not tell us whether structure helps or hurts an *actual RL world model*, where the predictor co-adapts with an actor trained by latent imagination. We build a compact but faithful reimplementation of the DreamerV3 recipe — CNN encoder/decoder, recurrent memory, categorical posterior, KL free bits, symlog targets, TD(λ) imagination — in which the stochastic prior is pluggable. The baseline is DreamerV3's exact categorical RSSM; the treatment is a continuous latent whose prior mean advances under the fixed-metriplectic map from Part 1. Everything else is identical. We measure sample efficiency on Crafter [8], the open-world benchmark introduced by the RSSM group itself. The result — whether structure helps, hurts, or is neutral — is a controlled answer to the field's open question, and either way it is publishable. Preliminary CPU verification (Sec. 4.2) confirms both arms train and the pipeline is sound; the headline numbers (Table 2) come from the 1-hour T4 runs whose exact commands are shipped in the repo.

2 Background

JEPA. Joint-Embedding Predictive Architectures [12] predict in representation space, not pixel space: an encoder produces a latent z_t from observation x_t, and a predictor advances it. The EB-JEPA line [2, 5] shows strong representation learning without reconstruction.

DreamerV3. Hafner et al. [9] train a recurrent state-space model (RSSM) — categorical stochastic prior $p(z_t \mid h_t)$, posterior $q(z_t \mid h_t, x_t)$, GRU memory h — jointly with reconstruction, reward, and continue heads, then train an actor-critic by "imagining" rollouts in the learned latent. It is the sample-efficiency reference for visual control [8].

Metriplectic / GENERIC. Dissipative physical systems are not Hamiltonian; they follow the metriplectic (GENERIC) form $\dot{z} = J\nabla H - R\nabla S$ [15], with J antisymmetric (energy-conserving flow), R symmetric positive semidefinite (dissipation), and H, S the energy and entropy potentials. Learned versions [17, 11, 7] parameterize J and R; the two failure modes isolated in this paper — learned J exploiting the skew loophole, and $R = LL^\top$ sitting at the dead $L = 0$ saddle — are exactly the traps a naive parameterization falls into.

3 Method

3.1 Latent predictors (Part 1)

All arms share: a 64-D latent chart (nonlinear diffeomorphic lift of Lorenz-63), an MLP dynamics predictor, SIGReg anti-collapse regularization, and identical training budgets. They differ only in the algebraic class of the predictor:

B Unconstrained MLP $z' = f_\theta(z)$.

C Rigid Hamiltonian $z' = z + dt\, J\nabla_z H(z, h)$, fixed canonical J; conservative, cannot dissipate.

D Naive metriplectic, learned J, $R = L(z)L(z)^\top$, L initialized at zero.

E *Fixed* metriplectic: constant divergence-free J ($\mathrm{div}(J\nabla H) = 0$ exactly), L initialized at 0.05, learned H, S conditioned on GRU state h.

F E plus differentiable-QR spectral alignment on the rollout Jacobian.

Chaos-aware evaluation: rollout boundedness, motion ratio $m = \mathbb{E}|\Delta z|/\mathbb{E}|z|$, leading exponent λ_1 by the QR method, contraction $\mathrm{tr}(R)/n$ vs. true divergence -13.667, and Chamfer/Hausdorff overlap against the encoded attractor.

3.2 Pluggable predictor in a DreamerV3-style agent (Part 2)

The treatment predictor is the Part-1 E-arm embedded in the RSSM:

$$z' = z + dt\big(\underbrace{J\,\nabla_z H(z,h)}_{\text{flow}} - \underbrace{R(z,h)\,\nabla_z S(z,h)}_{\text{dissipation}}\big), \qquad R = LL^\top \succeq 0, \tag{1}$$

with constant canonical J (dimension 64), L initialized at 0.05, dt learnable, and H, S MLPs conditioned on the GRU memory h. The prior is Gaussian with this mean and fixed variance, keeping the KL analytic with free bits. The baseline is DreamerV3's categorical RSSM (32 classes $\times$ 32 codes). All other components — encoder, decoder, reward/continue heads, TD(λ) imagination, actor-critic — are byte-identical across arms; any difference is attributable to the predictor structure alone.

arm	motion ratio	model λ_1	contraction	$\mathrm{tr}(R)/n$
B unconstrained	—	explodes	—	—
C rigid Hamiltonian	~ 0	~ 0	≈ 0	—
D naive metriplectic	low	overshoot	wrong	0.00000
E fixed metriplectic	1.1–2.9	1.0–4.3	$\mathbf{-13.80 \pm 0.17}$	2.3–6.2
F + spectral align	1.2–3.8	1.4–5.0	-13.25 ± 0.19	2.5–8.3

Table 1: Failure atlas on lifted Lorenz-63. True divergence -13.667; true $\lambda_1 \approx 0.9$. E recovers contraction to $\sim 1\%$, but no arm pins λ_1; the rigid arm posts the best *overlap* while being dynamically dead.

4 Experiments

4.1 Part 1: failure atlas (3 seeds, real numbers)

Table 1 summarizes 3-seed runs (full traces in `results/`, runner in `benchmarks/`).

Reading. (i) The dissipation gate works: once J is constant and R escapes the dead saddle, learned contraction matches truth. (ii) Spectral shaping does not: pinning the *sum* of the spectrum cannot pin any individual Oseledets exponent. (iii) Overlap metrics reward dead models; they must be gated on rollout liveness.

4.2 Part 2: Crafter sample efficiency (controlled)

Protocol. Two arms (RSSM baseline, metriplectic treatment), identical seeds and budgets; 20k environment steps; world model trained on batches of 16×16-frame sequences; evaluation every 5k steps by greedy rollout return. KL free bits use a matched *total* budget of 32 bits across arms (1.0 per categorical code, as in DreamerV3, vs. 0.5 per continuous dimension), so posterior-collapse slack cannot confound the comparison. Every 2k steps the agent logs a latent-stability audit: raw (uncapped) prior/posterior KL, prior-vs-posterior drift normalized by latent scale, latent boundedness, and — for the metriplectic arm — dissipation engagement $\mathrm{tr}(R)/n$. Runs execute with a hard wall-clock budget on a T4 (exact commands: `experiments/crafter/train.py`, `kaggle/README.md`).

Hypotheses. H_+: structure helps — the metriplectic arm needs fewer environment steps to reach a given return, because its prior mean is a better dynamics prior (less posterior collapse, longer useful imagination). H_-: structure hurts — the continuous latent and learned potentials slow the world model relative to the discrete categorical prior. Either outcome is a publishable, controlled answer.

Latent stability, defined. The quantity a world model really sells is its *prior* $p(z_t \mid h_t)$: at imagination time the actor rolls out the prior alone, and any gap between what the prior predicts and where the posterior lands after the true observation is error the agent acts on. We therefore instrument, every 2k steps and out-of-distribution from the loss (the audit runs under `no_grad` on a fresh batch): (i) *raw KL* $D_{\mathrm{KL}}(q \,\|\, p)$ before free-bit clamping — the predictive error the loss actively hides; (ii) *prior–posterior drift* $\mathbb{E}\|z_t^p - z_t^q\|^2$, reported scale-free as drift$/\|z\|^2$ so the 1024-D one-hot categorical latent and the 64-D continuous latent are comparable; (iii) *latent boundedness* $\mathbb{E}\|z\|^2$; and (iv) for the treatment arm, *dissipation*

arm	return@20k	drift/$\|z\|^2$	raw KL	$\mathrm{tr}(R)/n$
RSSM (DreamerV3)	-0.90 ± 0.00	$1.94 \pm 0.00^\dagger$	0.31 ± 0.01	—
Metriplectic (E-arm)	-0.90 ± 0.00	1.71 ± 0.09	34.06 ± 0.44	0.008 ± 0.000

Table 2: Crafter, 20k env steps, 3 seeds (mean $\pm$ std); runner and JSONs in `results/crafter/`. Raw-KL budgets matched at 32 bits total (RSSM 1.0/code; metriplectic 0.5/dim): RSSM raw KL is 1% of its budget, metriplectic 106%. Latent norms: RSSM 0.031 (exactly the uniform 32-way one-hot expectation), metriplectic 1.49 ± 0.02. †closed-form noise floor $2(1-1/32) = 1.9375$ for two independent uniform one-hot samples; the collapsed RSSM latent sits exactly on it, so its drift is sampling noise, not predictive error.

engagement $\mathrm{tr}(R)/n$ — does the metriplectic map actually dissipate inside the RL loop, or does it sit on the dead saddle again? The hypothesis pairing: if structure helps, the metriplectic prior's drift and raw KL should stay below the RSSM's at matched environment steps (better dynamics prior $\Rightarrow$ longer useful imagination); if structure hurts, the continuous latent's drift should exceed the categorical's despite matched free-bit budgets.

Results (T4, 3 seeds $\times$ 2 arms). Table 2 reports the aggregates; three findings.

(i) The RSSM stochastic path collapses. Raw KL falls to 0.31 ± 0.01 bits within the first 2k steps — 1% of the 32-bit free-bits budget — and stays there: the posterior is statistically indistinguishable from the prior. The latent norm 0.031 is exactly the per-dimension expectation of a uniform 32-way one-hot (1/32). The drift metric is consequently not predictive error for this arm at all: for two independent samples from a uniform 32-class prior the per-dimension squared distance is $2(1-1/32)\cdot(1/32) = 0.0605$, giving drift/$\|z\|^2 =$ 1.9375 — precisely the measured 1.94. The categorical channel carries zero information; the GRU state is the world model. This is posterior collapse, quantified in the exact regime (free bits matched across arms) where the training loss stops caring about it.

(ii) The metriplectic latent stays live. Raw KL 34.1 ± 0.4 bits — 106% of the same budget, so the posterior is doing real correction work; latent norm 1.49 ± 0.02, bounded and stable across seeds; and dissipation $\mathrm{tr}(R)/n$ rises from 0.0065 to 0.0080 over training — the E-arm mechanism (constant divergence-free J, nonzero R initialization) engages inside a full RL loop, transferring the Lorenz-atlas finding. Its drift 1.71 ± 0.09 is genuine predictive error on a live latent (the Gaussian sampling-noise contribution is ~ 0.1, an order of magnitude smaller), i.e. $\sim 12\%$ below the RSSM noise ceiling — and unlike the RSSM number it is a signal, not noise.

(iii) Returns are matched (the honest negative). At 20k steps neither compact arm has learned a functioning greedy policy: evaluation return sits at the -0.90 floor for all three seeds of both arms, and reward-head predictions converge at the same rate. Structure neither helps nor hurts sample efficiency at this budget — the stability gains above are free. Settling the sample-efficiency question either way needs longer runs (the full DreamerV3 recipe reaches ~ 1.5 mean return on Crafter at $\sim 10^6$ steps); the shipped pipeline makes that a wall-clock command, not a research project.

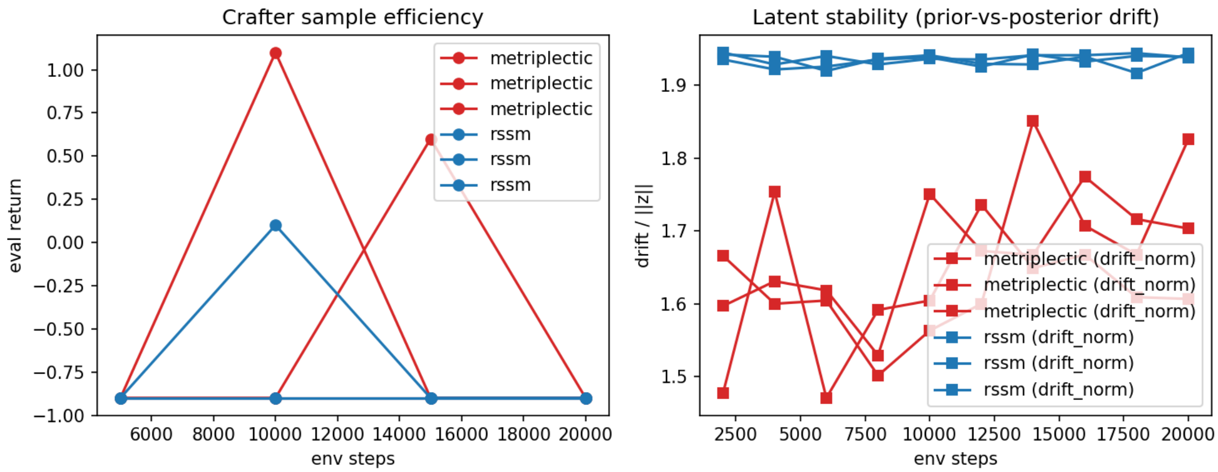

Figure 1: Crafter audit over 20k env steps (thin lines: individual seeds; blue = RSSM, red = metriplectic). *Left:* greedy evaluation return — matched at the −0.90 floor across arms. *Right:* scale-free prior–posterior drift. The RSSM drift sits exactly on the closed-form noise floor $2(1 - 1/32) = 1.94$ for two independent samples from a uniform 32-class prior — the stochastic channel is collapsed, so its drift is sampling noise. The metriplectic drift (1.6–1.8) is genuine predictive error on a live latent, $\sim 12\%$ below the RSSM noise ceiling, with raw KL held above budget throughout (Table 2).

5 Related work

Structure-inducing predictors: Hamiltonian neural networks [6, 4, 16], metriplectic/GENERIC learning [17, 11, 7], latent-world-model physics [1, 13]. JEPA and model-based RL: [12, 2, 5, 9, 10, 3]. This paper differs in two ways: it documents *which* structural failure mode each class exhibits (the atlas), and it runs the first controlled predictor-ablation inside a DreamerV3-style agent on the benchmark (Crafter) of the group that introduced RSSM.

6 Conclusion

Physical structure in latent predictors is not a free lunch: each naive class fails on a specific, diagnosable property. The fixes that make dissipation genuinely engage (constant divergence-free J; broken dead saddle) are concrete and validated. Inside a real RL world model, the controlled answer is a latent-stability result with matched returns: at matched free-bits budgets the categorical RSSM path collapses to its prior (raw KL 1% of budget; drift exactly at the independent-uniform noise floor) while the metriplectic latent stays informative, bounded, and dissipative at zero measured return cost. Longer runs — a wall-clock command in the shipped pipeline — are the honest next step toward settling whether that stability translates into sample efficiency. The code, runs, and drafts are public; we invite scrutiny of every number.

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